for which the solution subject to initial values $q(0) = q_0$, $p(0) = p_0$ is

$$\begin{pmatrix} q(t) \\ p(t) \end{pmatrix} = \mathscr{F}_t \begin{pmatrix} q_0 \\ p_0 \end{pmatrix} \equiv \begin{pmatrix} q_0 \cos(\Omega t) + p_0 \sin(\Omega t)/\Omega \\ -q_0 \Omega \sin(\Omega t) + p_0 \cos(\Omega t) \end{pmatrix}.$$

In somewhat more generality, suppose we are given an initial value problem for a linear system of ordinary differential equations,

$$\frac{\mathrm{d}\boldsymbol{z}}{\mathrm{d}t} = \boldsymbol{A}\boldsymbol{z}, \qquad \boldsymbol{z}(t_0) = \boldsymbol{\zeta},$$

where $\boldsymbol{A} = (a_{ij})$ is a constant $m \times m$ matrix. The solution of this linear system may be described in several ways. If $\boldsymbol{A}$ has a basis of eigenvectors $\boldsymbol{\eta}_i$, $i = 1, \ldots, m$, with corresponding eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_m$, then we may write the solution at time t as

$$\boldsymbol{z}(t) = \sum_{i=1}^{m} c_i e^{\lambda_i (t - t_0)} \boldsymbol{\eta}_i$$

where the coefficients c_i are obtained by solving the equation

$$\boldsymbol{z}(t_0) \equiv \boldsymbol{\zeta} = \sum_{i=1}^{m} c_i \boldsymbol{\eta}_i.$$

Observations: (i) eigenvectors, eigenvalues, and coefficients c_i may be complex, but if $\boldsymbol{A}$ and $\boldsymbol{\zeta}$ have real coefficients, it is nonetheless possible to obtain a real solution, and (ii) with our assumption that the $\{\boldsymbol{\eta}_i\}$ form a basis, the calculation of the coefficients is always possible, since the matrix $\boldsymbol{X}$ whose columns are the eigenvectors will be invertible, that is, the coefficients c_i can be enumerated as the components of a vector $\boldsymbol{c}$ which satisfies the square linear system

$$\boldsymbol{\zeta} = \boldsymbol{X}\boldsymbol{c}.$$

An alternative expression for the solution is in terms of the exponential of the matrix $\boldsymbol{A}$ scaled by time,

$$\boldsymbol{z}(t) = e^{\boldsymbol{A}(t - t_0)} \boldsymbol{\zeta}.$$

The exponential of a matrix can be defined, in the case of a matrix with a full set of eigenvectors, in terms of the eigenvector expansion. Alternatively, one may think of $\exp(\boldsymbol{A})$ as the sum of the exponential series

$$e^{\boldsymbol{A}} \equiv \boldsymbol{I} + \boldsymbol{A} + \frac{1}{2!}\boldsymbol{A}^2 + \frac{1}{3!}\boldsymbol{A}^3 + \ldots$$